

QUANTUM MECHANICS

Wave Nature of Matter

1 time for 3 Marks

Wave nature of matter is *not visible* in our daily life.

Particle of mass 'm' and velocity 'v' is associated with a wave of wavelength :

$$\lambda = h / mv = h / p$$

where,

p: momentum of the particle.

Not associated with:

Rest particle, $v = 0$ and $\lambda = \infty$

Macroscopic particle for which $m \gg 0$ and $\lambda \cong 0$.

De Broglie's Hypothesis

To Prove: $\lambda = h / \sqrt{2mE}$

4 times for 3 Marks

De Broglie introduced the concept of matter waves. He suggested that the wave particle duality of radiation should be extended to micro-particles as well.

1. From the *theory of radiations*: $E = h\nu = hc / \lambda$
2. Einstein's *mass energy relation*: $E = mc^2$
3. Equating (1) and (2): $hc / \lambda = mc^2$
4. Re-arranging (3): $h = p\lambda$
5. (Kinetic energy eqn) $E = mv^2 / 2 = (mv)^2 / 2m$
6. $E = p^2 / 2m$
7. Putting (4) in (6): $E = h^2 / 2m\lambda^2$
8. Thus the value of λ is given by: $\lambda = h / \sqrt{2mE}$

Wavelength of matter wave of an electron accelerated at potential V

To Prove: $\lambda_T = 1.67\text{\AA} \approx \lambda_E$

1 time for 3 Marks

1. **Electron energy:** $E = (mv^2 / 2) = eV$
2. **Wavelength** associated with an **electron wave:** $\lambda_T = h / \sqrt{2mE}$
3. Putting (1) in (2): $\lambda_T = h / \sqrt{2meV}$
4. Putting values for h, m, e in (3): $\lambda_T = 12.25 / \sqrt{V}$

- Importance of this wavelength: when $V = 54 \text{ volts}$, theoretical electron wavelength is $\lambda_T = 1.67\text{\AA} \approx \lambda_E$ (Experimental value)
- This agreement confirms de Broglie's hypothesis of matter waves.

PROPERTIES OF MATTER WAVES

2 times for 5 Marks

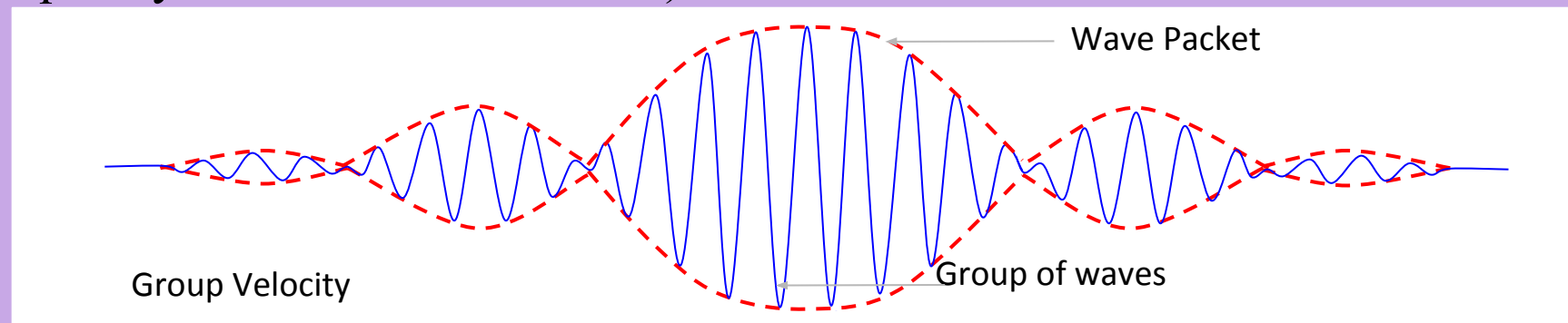
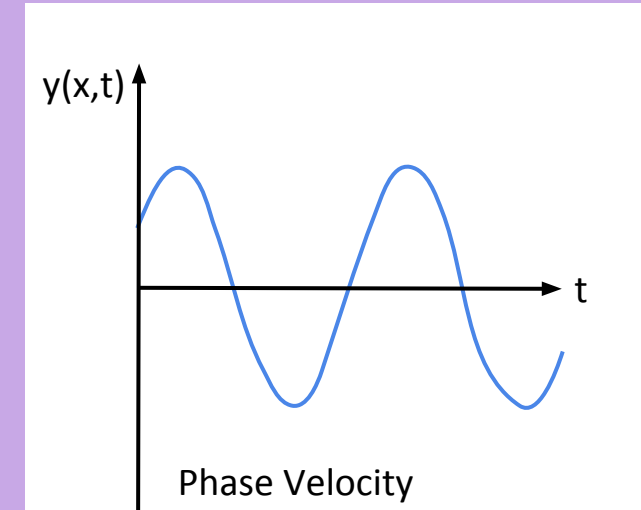
- Not mechanical nor electromagnetic waves, these are *hypothetical waves*.
- Matter waves with *different de Broglie wavelengths* travel with *different velocities*.
- The de Broglie wavelength depends on the *kinetic energy* of the matter particle.
- Matter waves travel *faster than light*.
- The energy of a matter wave being combined with Einstein's mass energy formula:
$$h\nu = mc^2$$
$$v = mc^2 / h$$
- Wave velocity of the matter:
$$u = v\lambda = mc^2\lambda / h$$
$$u = mc^2 h / hp$$
$$u = mc^2 / mv$$
$$u = c^2 / v$$
$$u = c(c / v) > 1$$
- *Since no matter particle can travel at a velocity greater than c, $v \ll c$ and thus $(c/v) \gg 1$.*

Wave Packet, Phase and Group Velocity

3 times for 5 Marks

- **WAVE PACKET:** A particle confined to an extremely small volume is represented by a combined version of multiple waves of slightly different frequencies.
- **PHASE VELOCITY:** The rate at which the phase of the wave changes at a point in the medium is called the phase velocity of the wave in that medium.(velocity of a monochromatic wave)
$$v_p = \omega / k$$
- **GROUP VELOCITY:** The rate at which the energy propagates through the medium while a group of waves within a frequency band is travelling through it is called the group velocity.(velocity with which a *group* of waves within a frequency band $d\omega$ would travel.)

$$v_g = d\omega / dk$$



Significance of the wave function Ψ of matter waves.

1 time for 3 Marks

- Particles exhibit *wave* properties.
- Provides information about the *probability of finding* a particular particle at a *given position* and at a *given time*.
- Helps in finding the *amplitude of position, momentum* and other *physical properties* of a matter particle.

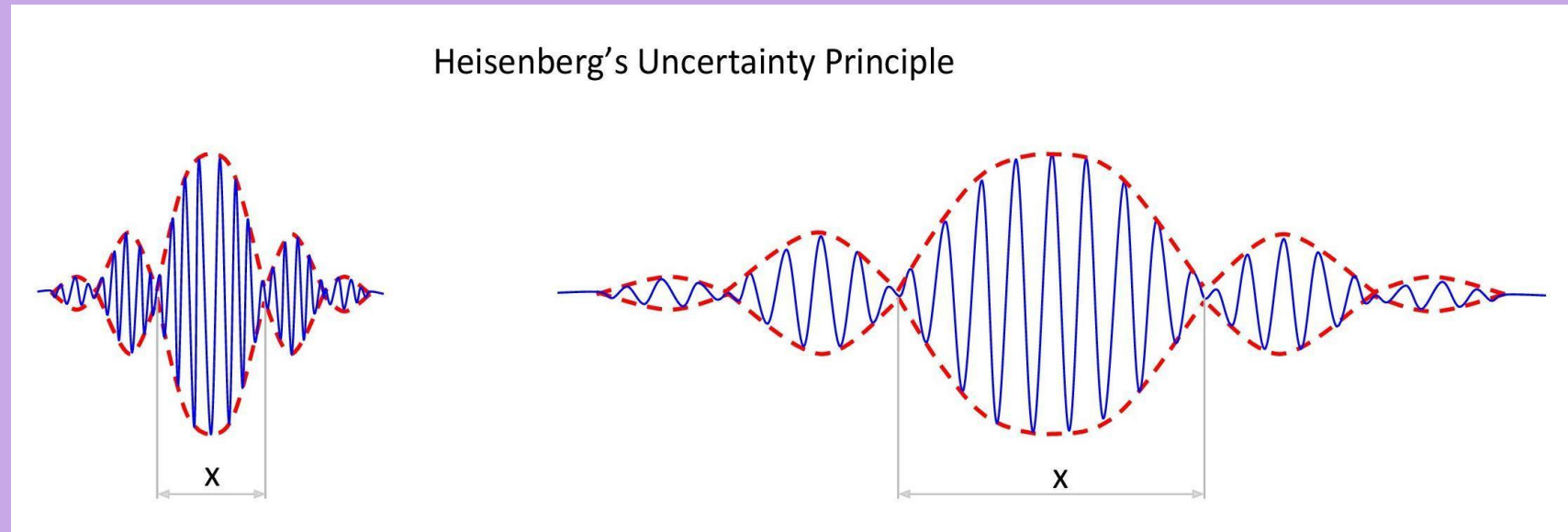
Heisenberg's uncertainty principle

4 times for 5 Marks

- The position and momentum of a particle cannot be determined precisely and simultaneously. This is stated in the Heisenberg's Uncertainty Principle as the inaccurate Δx and Δp in the simultaneous determination of the position 'x' and momentum, 'p' respectively of a particle are related as,

$$\Delta x \cdot \Delta p \geq \hbar$$

- It's physical significance is to determine and prove minimum energy possessed by an electron in the atom and the non existence electrons in a nucleus respectively.



Using Heisenberg's principle show that electrons cannot exist within the nucleus.

10 times for 5 Marks

1. Assuming the electron is a part of the nucleus.
2. Size of a nucleus: 1 fermi = 10^{-15} m.
3. Uncertainty in position: $\Delta x_{\max} = 10^{-15}$ m
4. Heisenberg's uncertainty principle: $\Delta x_{\min} \cdot \Delta p_{\min} = \hbar$
5. Substituting values: $\Delta p_{\min} = (\hbar / \Delta x_{\min}) = (6.63 \times 10^{-34}) / (2 \times 3.14 \times 10^{-15})$
6. $\Delta p_{\min} = (\hbar / \Delta x_{\min}) = 1.055 \times 10^{-19}$ kg-m/sec
7. $\Delta p_{\min} = m \Delta v_{\min}$

Using Heisenberg's principle show that electrons cannot exist within the nucleus. (Continued)

8. $\Delta v_{\min} = \Delta p_{\min} / m = 1.055 \times 10^{-19} / 9.1 \times 10^{-31} = 1.159 \times 10^{11} \text{ m/s}$
9. $v_{\min} < v, \quad v > 1.159 \times 10^{11} \text{ m/s} > c$
10. relativistic energy of electron: $E = \sqrt{(m_0^2 c^4 + p^2 c^2)}$
11. As $p \gg \Delta p_{\min}, p^2 c^2 \gg m_0^2 c^4$
12. Thus $E = pc$
13. Assuming $p = p_{\min}$: $E_{\min} = \Delta P_{\min} \cdot c$
14. Substituting values: $E_{\min} = 1.055 \times 10^{-19} \times 3 \times 10^8 = 3.165 \times 10^{-11} \text{ J}$
15. $E_{\min} = 3.165 \times 10^{-11} / 1.6 \times 10^{-19} = 197 \text{ MeV}$

Quantum Computing

- Focused on using the *principles of quantum theory* to develop various computing based technologies.
- In Quantum Computing, information is stored as an “*atom*” or a “*photon*” (a quantum system). A *qubit* (or a quantum bit) is the *fundamental unit of information*.
- The main benefits include much *less computation time* and *less bulky* computers.